Xixiao (Cecilia) Pan

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Education

- **University of Southern California** — Aug. 2025 - Expected May. 2027
  Master of Science in Computer Science, Viterbi School of Engineering — Los Angeles, California, USA
- **University of Michigan, Ann Arbor** — Sep. 2023 - May 2025
  Bachelor of Science in Engineering, Data Science Engineering, College of Engineering — Ann Arbor, Michigan, USA
- **UM-SJTU Joint Institute, Shanghai Jiao Tong University** — Sep. 2021 - May 2025
  Bachelor of Science in Engineering, Electrical and Computer Engineering — Shanghai, China

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, Database Systems, Distributed Systems, Machine Learning, Web Dev, Computer Vision, Information Retrieval

Skills

- **Programming Languages:** C++, Java, Golang, C#, JavaScript, Python, Unity, UE5, R, Matlab, SQL
- **Frameworks & Tools:** PyTorch, React, Flask, Redis, Docker, MongoDB, Agile, AWS, Kubernetes, Linux

Experience

- **Advanced Micro Devices, Inc (AMD) []** — May 2024 - Apr. 2025
  AI Developer/Software Developer Intern, Tech Stack: PyTorch, React, LLM, Django, Prompt Engineering — Shanghai, China/Remote
  - Created Wingman AI Debug Agent utilizing **React** and **Django**, interpreting natural language commands to automate driver workflows, resulting in **70%** increased development efficiency.
  - Combined AMD **LLM** API to decompose complex tasks into subtasks and boosted task execution accuracy by **30%** through **prompt engineering**, few-shot learning, and role-based message design.
  - Optimized Stable Diffusion 3 model by applying **multi-head attention** transformations, **ONNX** graph rewrites, and **transformer** structure improvements, resulting in **3×** inference speedups on AMD GPUs.
  - Reduced memory footprint and boosted throughput through AWQ **quantization** techniques and conversion of **PyTorch** models to ONNX format using Microsoft Olive for cross-platform deployment.
- **Artisk.ai []** — Mar. 2025 – Jun. 2025
  AI Developer Intern, Tech Stack: Python, Flask, Redis, Docker, JavaScript, OpenCV, Google Cloud — Remote, USA
  - Engineered a color conversion function API for an AI logo Gen product with AI agent interaction, deployed as a containerized **Flask** microservice with **Docker** on **GCP** supporting **10,000+** daily requests.
  - Implemented a **Redis** asynchronous task queue with request polling and real-time updates, enabling smooth client-side interaction via **JavaScript** while ensuring non-blocking image processing workflows.
- **GetStitch.ai, Startup Project []** — Feb. 2025 - Mar. 2025
  Student Contributor, Tech Stack: Python, FastAPI, Neo4j, TypeScript, React, AWS — Ann Arbor, Michigan, USA
  - Engineered an AI-powered learning web using **Claude 3 LLM** via **AWS Bedrock**, with custom prompt engineering and validation to automatically generate study materials from uploaded files.
  - Developed **FastAPI**-based **RESTful APIs** to facilitate CRUD operations for Neo4j-stored graph data with **Redis** caching, request batching, and selective hydration to optimize response speed.
- **Ping An Technology Co., Ltd.[]** — May 2023 - Aug. 2023
  Backend Development Intern, Tech Stack: Spring Boot, MyBatis, Quartz, RocketMQ, Apollo, MySQL — Remote, China
  - Led iBatis-to-**MyBatis** migration by developing an industry-standard Java-based transformation utility with regex pattern matching that reduced manual migration effort by **90%**.
  - Implemented **RocketMQ**-based messaging system with support for retry logic and Dead Letter Queues (**DLQ**) to ensure message reliability and failure recovery in core loan repayment processing workflows.

Projects

- **miProxy – HTTP Video Proxy** - *Python, Mininet, Networking* [] — Aug. 2024 – Dec. 2024
  - Developed adaptive **HTTP proxy** for MPEG-DASH, passing 20+ automated tests with **100%** coverage and sustaining **30%** higher average bitrate utilization under bandwidth fluctuations.
  - Simulated a **CDN** in Mininet with **6+** nodes and variable latency; built DNS-based **load balancer** ensuring <**5%** variance in server distribution and stable playback for 10+ concurrent browser streams.
- **Distributed key-value storage** - *Go, Distributed System* [] — Oct. 2024 - Dec. 2024
  - Built a fault-tolerant, **Paxos**-based distributed key/value store in **Go** using **RPC** for communication, supporting linearizable operations, at-most-once semantics, and concurrent consensus instances.
  - Extended the system with a **sharded** architecture and a Paxos-backed ShardMaster to enable dynamic reconfiguration, cross-group shard transfers via RPC, and horizontal scalability under network failures.
- **Search Engine** - *C++, GCP, System Design* [] — Jan. 2025 - May 2025
  - Designed a distributed search engine in C++ featuring **multithreaded** crawlers with **Bloomfilters**, fault-tolerant checkpointing, HTML parser, delta-encoded inverted index, and a TDRD query compiler.

Awards

- **Finalist, Grundfos Prize Student Award 2025 (Project Student/Group Category)** — Bjerringbro, Denmark
  - Recognized for AI-based predictive maintenance research at Grundfos Headquarters.